### 9.21.5 Purposive selection

In **T 513/90** (OJ 1994, 154) the board held that if, for a particular application of a known process, the skilled person could obviously use a material generally available on the market and suitable for the purpose, and was also highly likely to use it for reasons irrespective of its characteristics, such use should not be considered as inventive on account of those characteristics alone. It stood to reason that if carrying out such a step was itself already obvious for other reasons, the natural choice of the particular means on the market-place was devoid of mental or practical effort, or of "purposive selection", in the absence of anything to the contrary (see also **T 659/00**, **T 1861/16**).

In **T 636/09** the board held that no inventive step was entailed in accepting a lower yield likely when using a more readily available raw material (e.g. where industrial hemp is substituted for marijuana (cannabis), the latter being more readily available for legal reasons).

### 9.21.6 Automation

According to the established case law of the boards, the mere automation of functions previously performed by human operators is in line with the general trend in technology and thus could not be considered inventive (**T 775/90**, **T 1175/02**, **T 438/06**, **T 734/13**, **T 711/14**, **T 2315/16**).

The mere idea of executing process steps automatically, e.g. replacing manual operation by automatic operation, was a normal aim of the skilled person (**T 234/96**).

In developing an automated process from a known manual process, apart from simply automating the individual steps of the manual process, the skilled person will also incorporate the facilities that automation typically offers for the monitoring, control and regulation of the individual process steps, provided they fall within the definition of technical skill (**T 850/06**). See also **T 2315/16**.

### 9.21.7 Enhanced effect

According to the case law of the boards of appeal enhanced effects could not be adduced as evidence of inventive step if they emerged from obvious tests (**T 296/87**, OJ 1990, 195; **T 432/98**; **T 926/00**; **T 393/01**).

In **T 308/99** the claimed use was based on a thoroughly obvious property of known substances. The slightly enhanced effects associated with the claimed use in comparison with substances used in prior art emerged from obvious tests.

In **T 104/92** the board held that work involving mere routine experiments, such as merely conventional **trial-and-error** experimentation without employing skills beyond common general knowledge, lacked inventive step. It would be obvious for a skilled person to use varying proportions of known polymers for outer layers with a reasonable expectation of obtaining better shrink or shrink and heat-seal characteristics for the laminate of the invention. See also in this chapter I.D.7.2 "Try and see situation".

In **T 253/92** the subject-matter of claim 1 related to a process for the manufacture of a permanent-magnet alloy. In the board's view, a skilled person would have regarded it as obvious to try out a variety of alloys known from the prior art to be of similar composition to those of the better examples and to measure their magnetic properties.

In **T 423/09** the board stated that the enhanced effect did not emerge from routine tests but from the practice to be followed according to the rules and recommendations of the handbook. The skilled person following the recommended practice prescribed in this handbook, and thus acting only routinely would inevitably obtain this enhanced effect, which therefore could not be taken as an indication of inventive step.

In **T 237/15** the technical problem was the provision of a treatment regimen for human patients based on oral administration of suberoylanilide hydroxamic acid. The board found that the determination of the optimum dosage regimen required to achieve the therapeutic effect in the (human) patient was a matter of routine experimentation for the skilled person. It held that such routine tests did not require inventive skill and could consequently not establish an inventive step. See also in this Chapter I.D.9.21.12 "Animal testing and human clinical trials".

### 9.21.8 Simplification of complicated technology

In **T 61/88** the board indicated that, in the face of an optimal but sophisticated solution to a technical problem, the skilled person could not be denied the capacity to recognise that less complicated alternatives generally achieved less perfect results and consequently to envisage such alternatives, at least in situations in which the advantages of decreased complexity could reasonably be expected to outweigh the resulting loss of performance (**T 817/94**).

In **T 505/96** the board concluded that the simplification of complicated technology in situations in which the advantages of decreased complexity could reasonably be expected to outweigh the resulting loss of performance must be considered to be part of the normal work of the person skilled in the art.

### 9.21.9 Choice of one of several obvious solutions

a) Arbitrary choice from a host of possible solutions

A merely arbitrary choice from a host of possible solutions cannot be considered inventive if not justified by a hitherto unknown technical effect that distinguishes the claimed solution from the other solutions (**T 939/92**, OJ 1996, 309; **T 739/08**; **T 1175/14**; **T 2554/16**; **T 1448/15**; **T 115/18**; **T 1984/15**; **T 1862/15**). In **T 400/98** the board stated that applying one of the possible solutions which were available to the skilled person required no particular skills and hence did not involve an inventive step (see also **T 107/02**). In

**T 1984/15** the board held that both single and repeated arbitrary choices were part of the skilled person's routine approach to providing an alternative solution.

In **T 1862/15** the board held that in a case involving an arbitrary selection, the prior art did not need to contain an incentive for the skilled person to select the particular solution claimed. Instead, all possible solutions had to be regarded as being equally suitable and obvious candidates for solving the objective technical problem; therefore, they all had to be considered to be suggested to the skilled person.

In **T 588/99** the board stated that in the particular situation where a document explicitly defined any compound having a certain activity as a suitable component of a detergent composition, and urged the skilled person to look for such compounds in publications of other technical fields such as biochemistry and medicine, it required no inventive activity to solve the technical problem of providing an alternative to the compositions disclosed in such prior art by replacing the explicitly specified compounds having the given activity with any other such compounds which may be found by exploring the other technical fields.

In **T 1941/12** the board stated inter alia that an arbitrary selection of strains, **by the very fact of it being arbitrary**, did not involve any inventive step. This was even more so since the two specific strains chosen were known from the prior art as commercially available.

In **T 892/08** the board referred to the established case law whereby, when the technical problem is simply that of providing a further composition of matter or a further method, i.e. an alternative to the prior art, any feature or combination of features already conventional for that sort of composition of matter or method represented an equally suggested or obvious solution to the posed problem. The boards have repeatedly established that the simple act of arbitrarily selecting one among equally obvious alternative variations is devoid of any inventive character. See also **T 311/95**, **T 89/16**.

In **T 816/16** the board held that the further addition of components well known for their activity in the technical field concerned was considered an arbitrary choice among components which the skilled person would have had at their disposal.

b) Selection from obvious alternatives

In **T 190/03** of 29 March 2006 the board stated that in connection with the obviousness of a solution chosen from various possibilities, it was sufficient that the one chosen was obvious and not necessarily relevant that there were several other possible solutions (see also **T 214/01**, **T 688/14**).

In **T 357/02** the board found it followed from the minimalist character of the technical problem objectively arising from the closest prior art, which could only be formulated as a modification of that state of the art, that regardless of a success or failure of the measures applied, almost any modification of the latter process might be regarded as a feasible alternative by the person skilled in the relevant art, and therefore obvious, since each corresponding solution would be equally useful (or useless).

In **T 1072/07** the board concluded that to solve the problem (how to select a suitable type of burner), the person skilled in the art had to make a choice between two well-known possibilities. Either choice, which in a particular situation would be based on balancing the advantages of the specific type of burner being selected, such as efficiency in its operation, with its disadvantages, such as technical adaptations required and costs involved, was therefore obvious and did not involve an inventive step.

In **T 1045/12** the appellant (applicant) argued that the solution to the objective technical problem taught by D3 was one of several, equally likely options and that the board had to provide a reason why the skilled person would have selected the claimed option. The board disagreed. The fact that there were other options had no bearing on the obviousness of one specific option. Furthermore, if all options were equally likely, then the invention merely resulted in an obvious and consequently non-inventive selection among a number of known possibilities. See also **T 288/15**.

9.21.10 Several obvious steps

If the technical problem that the skilled person has set himself to solve brings him to the solution step by step, with each individual step being obvious to him in terms of what he has achieved so far and what remains for him to do, the solution is obvious to the skilled person on the basis of the prior art, even if two or more such steps are required, and it does not involve an inventive step (**T 623/97**, **T 911/98**, **T 558/00**, **T 1514/05**).

9.21.11 Putting the closest prior art device into practice

In **T 405/13** the invention was a meter with rapid response glucose sensor further adapted to cause the measurement of current to occur at a time 5 seconds or less after the detection of sample application. The objective technical problem to be solved was to implement a timing circuit having a concrete point in time at which the timing circuit causes the measurement of current to occur. Deciding whether the claimed solution to this problem was obvious or not boiled down to the question of whether or not the skilled person, while putting the meter of D10 (biosensing meter) into practice, would arrive at a value falling within the claimed range (for a similar approach see **T 408/12** and **T 315/97** of 21 June 2002). In the board's view, no inventive step could be seen in including in the reasonably broad time range as required by the timing circuit of the meter of D10 the claimed value of 5 seconds, the less so since reducing measurement time had to be considered as an obvious desire of the user.

9.21.12 Animal testing and human clinical trials

In **T 1493/09** the board found that the skilled person seeking to solve the problem formulated (provision of a broadly effective vaccine against HPV, especially providing broad protection against cervical cancer) would have considered not only formulations for immediate use in human clinical trials but also additional animal testing. It took the view that its assessment according to the problem and solution approach applied for the same reasons to claim 1 of auxiliary request 1 (directed to the second medical use of the vaccine composition and including the therapeutic effect "prevention or treatment of a disorder related to HPV infection" as an explicit feature), and concluded accordingly that it too lacked inventive step. See also **T 237/15**.

In **T 237/15** the invention aimed to provide suitable dosages and dosing schedules of histone deacetylase inhibitors, especially suberoylanilide hydroxamic acid (SAHA), and to develop preferably oral formulations. The technical problem was the provision of a treatment regimen for human patients based on oral administration of SAHA. The board found the skilled person, in the knowledge (disclosed in D2) that SAHA was bioavailable when given orally in animal studies and having been given the information that SAHA achieved effective treatment in humans when introduced directly into the blood stream, would expect an effective treatment also for oral administration in human patients. The determination of the optimum dosage regimen required to achieve the therapeutic effect in the (human) patient was a matter of routine experimentation for the skilled person. The board held such routine tests did not require inventive skill and could consequently not establish an inventive step. See also in this Chapter I.D.9.21.7 "Enhanced effect".

## 10. Secondary indicia in the assessment of inventive step

### 10.1. General issues

According to established case law of the boards of appeal, a mere investigation for indications of the presence of inventive step is no substitute for the technically skilled assessment of the invention vis-à-vis the state of the art pursuant to Art. 56 EPC. Where such indications are present, the overall picture of the state of the art and consideration of all significant factors may show that inventive step is involved but this need not necessarily always be the case (see **T 24/81**, OJ 1983, 133 and **T 55/86**). Secondary indicia of this kind are only of importance in cases of doubt, i.e. when objective evaluation of the prior art teachings has yet to provide a clear picture (**T 645/94**, **T 284/96**, **T 71/98**, **T 323/99**, **T 877/99**). Indicia are merely **auxiliary considerations** in the assessment of inventive step (**T 1072/92**, **T 351/93**, **T 179/18**).

In **T 754/89** – "EPILADY" the board detailed its reasons for ruling that an inventive step was involved. Although factors such as commercial success, the overcoming of prejudice, the age of the documents cited, the cost of advertising and the creation of a new market segment, the satisfaction of a long-standing need, the existence of imitations and forms of infringement had received considerable attention, particularly in the parties' written submissions, the technical facts of the case were such that secondary indications of inventive step had lost any relevance.

In **T 915/00** the board held that commercial implementation, licensing and the recognition of the inventor's merits by the scientific community constituted further convincing secondary indicia for the presence of inventive step.

In **T 1892/12** the board observed that the boards had in some cases taken account of "secondary indicia" that an inventive step was involved but had not so far done so for the purposes of establishing that it was lacking. According to Art. 56 EPC, an invention shall be considered as involving an inventive step if, having regard to the state of the art, it is